

# PHC Pulse — AI/ML Specification

## Filled · HOW does the intelligence work?

**Purpose.** Define each AI component, its inputs/outputs, evaluation and fallback behavior.

*Filled from the repository PHC-Pulse-GDG-Code-for-community (main @ fd42d41) on 26 Sep 2026. Numbers quoted were measured by running the engine and the 13-test pytest suite (all passing) with PHC\_TODAY=2026-09-25, seed 42.*

Status legend: **[IMPLEMENTED]** in the repo and tested **[PARTIAL]** partly built **[PLANNED]** designed here, not yet in code

## 1. AI Components

| Component                       | Purpose                                                                  | Input                                                                                                    | Output                                                                                                                      | Method                                                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stockout forecast</b>        | Predict daily demand and when stock runs out before the next indent      | demand (dispensed + unmet) per PHC × drug × day; closing stock; supply_anchor; pipeline transfers; surge | daily_forecast, cover_days, days_to_stockout, p_stockout, status, surplus_units                                             | Damped Holt-Winters, multiplicative weekly seasonality ( $\alpha$ 0.25, $\beta$ 0.04, $\gamma$ 0.15, $\phi$ 0.9), vectorised over 1,200 series. Prod: BQML ARIMA_PLUS / Vertex AutoML |
| <b>Demand anomaly</b>           | Early outbreak signal and demand-shock adjustment                        | footfall_daily fever, diarrhoea, respiratory, OPD; PHC coordinates                                       | anomalies (observed, expected, ratio, z, consecutive days, severity), clusters, surge factor for linked drugs               | Weekday-adjusted robust z (median/MAD, 28-day baseline, Poisson floor); $z \geq 3$ , $\geq 1.5\times$ and $\geq 5$ excess; $\geq 2$ PHCs $\leq 50$ km = cluster                       |
| <b>Redistribution optimizer</b> | Choose donor → recipient shipments at minimum total cost                 | deficits (need, days_to_stockout), donors (surplus), road-km, cost parameters                            | shipments (lines, qty, km, INR, ETA, urgency), escalations, coverage stats                                                  | MILP via HiGHS (scipy.optimize.milp), per state, 8 s limit, 1% gap                                                                                                                    |
| <b>Voice extraction</b>         | Turn a spoken or typed report in any of 8 languages into structured data | 16 kHz WAV ( $\leq 60$ s) or text, PHC context, drug catalogue with local names, earlier partial report  | ParsedReport: transcript, language, translation, stock[], counts, beds, staff, confirmation, follow_up_question, confidence | Gemini 2.5 Flash multimodal with response_schema, temperature 0.1. Fallback: multilingual rule parser (nlu.py)                                                                        |
| <b>Capacity prediction</b>      | Bed and staff pressure per PHC                                           | beds_occupied / beds_total, staff_present / sanctioned, outbreak status                                  | <b>[PARTIAL]</b> occupancy and attendance ratios feed the resilience score (0–100, green/amber/red)                         | Rules today. Planned: Holt-Winters on occupancy + referral to the nearest facility with free beds                                                                                     |

## 2. Stockout Logic

*The template example (effective\_stock = current\_stock - safety\_stock; days = effective\_stock / demand) is replaced by the implemented, forecast-cumulative version:*

```

avail          = closing_stock(last count) + incoming(approved|in_transit) - committed(approved, donor side)
fc_adj[h]      = HW_forecast[h] * (1 + (surge - 1) * decay[h])    # decay = 1 until day 14, fades to 0 by day 28
days_to_stockout = first h where cumsum(fc_adj[1..h]) >= avail - days_since_last_count  (inf if never in 30 d)
next_supply_in  = days until next monthly indent (supply_anchor, 30-day cycle)
p_stockout      = P(cum demand to next supply > avail), Normal(mean = cumsum fc_adj, sd = sigma_28d * sqrt(h))
risk_level      = stocked_out if avail <= 0
                  critical   if days_to_stockout <= 7 and days_to_stockout <= next_supply_in
                  high       if days_to_stockout <= 14 and days_to_stockout <= next_supply_in
                  watch      if days_to_stockout <= 21 and days_to_stockout <= next_supply_in
                  surplus     if ok and surplus_units > max(10 * daily, 5)
                  ok          otherwise
surplus_units   = max(avail - (1.2 * cum_to_supply + 7 * daily) - 30 * daily, 0)    # safety factor 1.2, 7 safety days

```

- **Inputs:** 180 days of demand, the last closing balance, supply calendar, transfer pipeline and the anomaly surge (fever → PCM, ACT; diarrhoea → ORS, ZNC; respiratory → AMX).
- **Forecast method:** one recursive pass gives the model state at every past day. The forecast starts from each series' last observed day. Horizon 28 + 2 days.
- **Output schema:** {phc\_id, drug\_code, stock, incoming, committed, daily\_forecast, cover\_days, days\_to\_stockout, next\_supply\_in, p\_stockout, status, surplus\_units, surge, surge\_reason}.
- **Confidence:** p\_stockout (0–1) plus residual sigma over 28 days. A stock-out that lands after the next delivery is not flagged, which cuts false alarms.

### 3. Redistribution Objective

```

min  Σ_lanes F_p·y_p  +  Σ 0.5·x_pj  +  Σ deficits P_j · w_i · u_ij
F_p  = 600 + 18 · road_km (+ 2,500 if inter-state)      INR per trip; road_km = haversine × 1.3
w_i  = 1 + 14 / (1 + days_to_stockout)                urgency: 15× at day 0, ~2.4× at day 9
P_j  = PCM 60, ORS 45, ZNC 60, AMX 90, IFA 40, MET 70, AML 70, ACT 450, ASV 4000, OXY 900  (INR per unmet unit)
s.t. Σ_p x_pj + u_ij = need_ij                        need = forecast to (next supply + 7 safety days) - avail
      Σ_i x_kj ≤ surplus_kj                          donors keep 1.2× own forecast + 7 safety days + 30 days cover
      x_pj ≤ min(need, surplus) · y_p                 one y per donor→recipient lane shared by all drugs (consolidation)
      x, u ≥ 0, y {0,1}; 6 nearest donors per deficit; ≤ 220 km same state, ≤ 450 km cross-state

```

- Solved per state (the problem separates without cross-state lanes); national can enable cross-state.
- Leftover  $u \geq 1$  becomes an **emergency indent** escalation to the district warehouse.
- Expiry (planned): add  $-E \cdot x$  for lots expiring before the donor can use them, and restrict recipients to those whose forecast use before expiry  $\geq$  qty.

### 4. Human-in-the-Loop

| Condition                | Behavior                                                                                                                                                                                                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>High confidence</b>   | Voice: Gemini confidence $\geq 0.8$ and no follow-up. The read-back is shown and spoken; the worker taps Submit. Forecast: $p\_stockout \geq 0.8 \rightarrow$ shown as critical/high with a ready recommendation, still waiting for approval.                                                                                         |
| <b>Medium confidence</b> | Voice: 0.5–0.8, or a key field missing. Gemini asks one follow-up question in the worker's language and merges the answer; the worker reviews the fields before submitting. Rule-parser output is fixed at 0.6 and tagged <i>rules-fallback</i> .                                                                                     |
| <b>Low confidence</b>    | Voice $< 0.5$ or nothing parsed: do not submit. Ask the worker to repeat or switch to text/USSD. Forecast series with short history or a stale count (PHC not reporting) are shown as "watch" and reduce the resilience score, not auto-escalated. The confidence % is shown in the UI; the 0.5/0.8 gates are to be enforced in code. |
| <b>Critical action</b>   | Human approval required: <b>every</b> stock transfer (approve $\rightarrow$ dispatch $\rightarrow$ delivered, each a manual step by an officer in scope; PHC staff cannot approve), emergency indents, and IDSP outbreak alerts (Gemini drafts only). Stale recommendations are rejected (409).                                       |

### 5. Evaluation

| Component        | Metric                                                                                                               | Baseline                                                                                                                                                                                                                                                                  | Target                                                                                                               |
|------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Forecast</b>  | MAE/RMSE/MAPE: daily wMAPE on the 28-day hold-out, weekly wMAPE, stock-out recall $\geq 14$ days ahead and precision | Measured now: daily wMAPE 0.372; weekly wMAPE 0.21 (79% accuracy); recall 81.4% (302/371); precision 67.1%. Naive "last month's consumption" indent = 0% advance warning                                                                                                  | Weekly wMAPE $\leq 0.20$ ; recall $\geq 80\%$ ; precision $\geq 70\%$ on real pilot data                             |
| <b>Anomaly</b>   | Precision/Recall of clusters vs injected outbreaks (scenario truth)                                                  | Measured: clusters 2/2 found, 0 false (Koraput diarrhoea, 4 PHCs, up to 4.84×; Gorakhpur fever, 3 PHCs, 2.94×). PHC level: recall 8/8 injected (incl. lone Kalaburagi PHC-51 respiratory, correctly not clustered); precision 8/10 (PHC-99, PHC-118 are low-z, z 3.2–3.3) | Recall 100% of injected clusters, $\leq 1$ false cluster / district / month; detection $\leq 3$ days from onset      |
| <b>Optimizer</b> | Cost / stockouts prevented: INR per unit moved, coverage of need, stock-outs averted                                 | National plan: 118 shipments, 423 lanes, 5,578 of 31,505 units covered (17.7%), INR 198,431; 146 escalations. Deck lane: 284 strips for INR 1,347                                                                                                                         | Every same-state need with a donor $\leq 220$ km met; cost per averted stock-out day below the emergency-indent cost |

### 6. Failure / Fallback

- API failure (Gemini):** an exception is logged. Text reports fall back to `nlu.parse` with template read-back and follow-up. Audio-only returns HTTP 503 with a clear message (the UI offers browser STT or text). Briefings and IDSP alerts use deterministic templates. The copilot states it is unavailable. Every output is tagged with its engine.
- Missing data:** NaN demand keeps the model state and adds no residual. A PHC that has not reported is projected from its last count (elapsed days subtracted) and flagged. A series with no stock row starts at 0.
- Low confidence:** see section 4. Recommendations are never auto-executed.
- Optimizer failure:** 8 s time limit and 1% gap. If infeasible or empty, the plan returns no shipments with the solver message; shortages still show on the Stock page and as escalations.

- **Offline behaviour:** USSD, SMS and IVR need no data connection. The web report can be typed. Planned: an offline queue (IndexedDB) that resubmits voice/text when the connection returns, and an SMS fallback for officers' approvals.